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Smart Charging and Vehicle-to-Grid Integration for Sustainable Energy Systems: A Critical Review of Technical Performance and Deployment Readiness

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Abstract: Smart charging and vehicle-to-grid (V2G) integration can support renewable-energy utilization and power-system flexibility, but their practical contribution depends on more than the technical capability of individual vehicles and chargers. This review examines the conditions under which demonstrated charging and V2G performance can be translated into dependable services for sustainable energy systems. A critical integrative review was conducted using literature published between 2013 and August 2026. Studies were examined across battery, charger, distribution-network, communication, control, market, regulatory, and user dimensions, with particular attention to operating assumptions, performance baselines, and validation maturity. The reviewed simulations reported a 23.81% reduction in peak load at a photovoltaic (PV)–storage charging station, a 21% reduction in peak demand accompanied by an 18% increase in renewable-energy utilization in a microgrid, and a decrease in total harmonic distortion (THD) from 6.8% to 4.6% in a distribution feeder. These outcomes were obtained under different operating conditions and could not be directly compared. The synthesis identified persistent gaps between component-level performance and dependable deployment, particularly in fleet availability, battery degradation, interoperability, communication reliability, market participation, and field validation. A two-dimensional assessment framework was developed to distinguish installed, connected, controllable, and contractually dependable fleet capability while linking deployment decisions to successive levels of validation evidence. The findings indicate that reliable V2G deployment requires coordinated assessment of technical performance, operational constraints, and participant costs. The proposed framework provides a basis for evaluating deployment readiness and identifying the evidence needed to integrate electric vehicle (EV) flexibility into sustainable energy systems.

Keywords: Smart charging; Vehicle-to-grid; Sustainable energy systems; Renewable energy integration; Grid flexibility; Deployment readiness

1 Introduction

1.1 Highlights

- Four capability levels distinguish installed capacity from dependable V2G service.
- A coupled framework links fleet qualification with staged evidence maturity.
- Reported V2G benefits depend on system boundaries, baselines and assumptions.
- Field evidence remains limited for interoperability, secure fallback and settlement.
- Scalable V2G needs battery-aware control, mobility guarantees and fair compensation.

1.2 Review Scope, Objectives, and Analytical Approach

Fossil fuels have accounted for approximately 80% of global primary energy demand over recent decades [1]. However, the energy system is gradually transitioning toward cleaner and more diversified resources [1, 2]. This transition is reinforced by the Paris Agreement's objective of limiting the increase in global temperature to well below 2 °C [3]. Within the transportation sector, the adoption of vehicle-to-grid (V2G)-enabled electric vehicles (EVs) depends not only on technological capability but also on public perceptions of their potential benefits, risks, and practical feasibility [4]. V2G-based nanogrids have also been investigated as a means of supporting energy and environmental sustainability in rural areas [5]. These findings position EV–grid integration as both a technological development and a broader energy-transition challenge.

EV adoption has increased the need for well-planned, accessible charging infrastructure. Charging-station planning must account for spatial charging demand, user behavior, infrastructure location, and the interaction between vehicles and distribution networks [6]. At the same time, the sustainability of EV adoption depends on the environmental, infrastructural, and energy-system conditions under which electrification occurs [7]. Charging accessibility also varies between residential and workplace settings, indicating that infrastructure availability and location can influence the practical use of EVs [8]. The growing integration of EVs with renewable-energy systems further requires coordinated charging-station design, suitable control strategies, and effective energy management [9]. These developments demonstrate that transport electrification cannot be evaluated solely by vehicle adoption rates; its system-level value also depends on charging accessibility, infrastructure planning, renewable-energy coordination, and power-network capability.

Unmanaged EV charging introduces uncertain and time-varying electrical demand, making accurate charging-load forecasting increasingly important for power-system planning and operation [10]. The timing and duration of charging are influenced not only by technical conditions but also by users' capabilities, opportunities, motivations, and willingness to participate in smart-charging programs [11]. When charging demand is concentrated and uncoordinated, it may contribute to peak-load growth, voltage variation, transformer stress, congestion, and additional distribution-network losses [12]. However, the severity of these effects depends on network configuration, EV penetration, charger power, spatial concentration, charging schedules, and user behavior. Model-predictive control has been investigated as a means of coordinating charging stations in grid-connected microgrids and managing charging decisions according to system conditions [13]. The evidence therefore supports a transition from uncontrolled charging toward coordinated strategies, but the benefits of any control method must be interpreted within its specific network, demand, and validation conditions.

Smart charging coordinates EV demand based on grid conditions, electricity prices, renewable energy availability, and mobility requirements. In a charging-station case integrating photovoltaic (PV) generation, stationary storage, and EVs, V2G-enhanced optimization reduced peak load by 23.81% relative to disordered charging [14]. More broadly, V2G extends managed charging through bidirectional power exchange, whereas vehicle-to-everything applications also encompass energy exchange with homes, buildings, and external electrical loads. Aggregated EV fleets may provide temporal flexibility because many vehicles remain parked for much of the day [15], while multi-time-scale forecasting can estimate schedulable fleet capacity [16]. Nevertheless, practical flexibility is limited by vehicle connection status, bidirectional charger compatibility, battery condition, network constraints, forecast error, and mobility requirements.

Advanced computational methods are increasingly being applied to coordinate EV charging under coupled grid, operational, and user constraints. A federated reinforcement-learning framework that combines constrained Markov decision processes and graph neural networks reported a 13.6% reduction in grid operating cost while addressing network constraints and fairness among participating EVs [17]. For a renewable-supported microgrid, a multi-agent framework combining adversarial reinforcement learning (RL) with dynamic grey wolf optimization (DGWO) reported a 21% reduction in peak demand and an 18% increase in renewable-energy utilization [18]. A broader review also identified digital-twin simulations reporting a 63.5% reduction in system operating cost [19]. However, these numerical results are not directly comparable because they were obtained using different system models, datasets, baselines, objective functions, and operating assumptions. They demonstrate the potential of intelligent coordination under specific evaluation conditions but do not, by themselves, establish robustness, scalability, interoperability, or field readiness.

Battery degradation creates a fundamental trade-off between grid service provision and vehicle owners' interests. Repeated V2G cycling can contribute to both capacity and power fade: solid-electrolyte-interphase growth reduces usable capacity, whereas electrode degradation can increase internal resistance and reduce power capability [20]. Evidence from V2G service simulations further indicates that degradation is strongly influenced by energy throughput, charging regime, battery capacity, and, in particular, the depth of discharge used for ancillary services [21]. Battery-management strategies may limit the operating state-of-charge range to reduce exposure to conditions associated with accelerated aging. The appropriate limits depend on battery chemistry, temperature, service requirements, and the intended operating strategy [22]. Second-life use provides an additional pathway: batteries retired from

automotive service with approximately 60%–80% of their original capacity may remain suitable for stationary storage. A European Union material-flow analysis estimated that reusing around 40% of EV batteries for second-life applications could meet stationary-storage needs by 2040 while reducing cumulative primary demand for battery materials by up to 1.5% over 2020–2050 [23]. These findings indicate that the value of V2G cannot be assessed solely from grid-service revenue; it depends on battery chemistry, temperature, state-of-charge range, cycling intensity, mobility requirements, degradation costs, and the feasibility of second-life deployment.

Existing reviews and recent studies have examined smart charging, charging infrastructure, renewable-energy integration, microgrid control, market viability, and the technical, economic, regulatory, and social dimensions of V2G [14, 19, 24–26]. However, researchers often examine these domains separately rather than treating them as interdependent layers of an operational system. Consequently, a technically efficient charger or control algorithm may appear deployment-ready even when its reported performance depends on optimistic assumptions regarding vehicle availability, battery degradation, communication reliability, market access, or user participation. Furthermore, studies often compare performance improvements without adequately accounting for differences in network topology, EV penetration, baseline strategies, optimization objectives, mobility assumptions, and validation environments.

Several previous reviews have addressed specific technical and operational dimensions of V2G. Bibak and Tekiner-Moğulkoç [27] provide a comprehensive analysis of V2G systems and their applications, establishing a broad understanding of the technology's development and potential. Rana et al. [28] specifically examine bidirectional EV charger topologies and their associated control techniques, highlighting the power-electronic configurations underpinning V2G operation. More recently, Naseem et al. [29] examine smart charging and V2G integration, covering technical developments, cybersecurity risks, and mobility-oriented control strategies. Collectively, these contributions provide valuable insights into V2G architecture, bidirectional power conversion, control approaches, and operational challenges. Nevertheless, an important analytical question remains: under what operational conditions can the theoretical and technically demonstrated flexibility of EV fleets be reliably delivered as a grid service, and what level of evidence is necessary to substantiate such capability?

To address this gap, this review introduces an integrated framework that combines operational qualification and evidence maturity to assess V2G deployment readiness. The first dimension, operational qualification, distinguishes four nested levels of aggregate EV fleet capability: installed, connected, controllable, and contractually dependable. This distinction recognizes that installed battery capacity does not necessarily translate into flexibility available for grid services. Actual service delivery depends on vehicle connection, bidirectional charger compatibility, battery condition, state-of-charge constraints, mobility requirements, network capacity, communication reliability, market eligibility, and user participation. The second dimension, evidence maturity, evaluates the credibility of deployment claims through successive validation stages, progressing from reproducible modeling and laboratory testing to hardware-in-the-loop (HIL) validation, integrated pilots, network-scale demonstrations, and system readiness. By linking these dimensions, the framework moves beyond the separate assessment of technical capability and validation maturity to examine whether operationally qualified fleet flexibility is supported by sufficient evidence for dependable service delivery. It therefore provides a structured basis for distinguishing nominal or technically demonstrated V2G potential from flexibility that can be credibly committed, delivered, and verified at progressively larger deployment scales.

Building on this framework, the review examines smart charging and V2G as an interconnected battery–charger–grid–communication–market–user system. It critically evaluates how charging architectures, power-electronic interfaces, control strategies, and operational objectives interact with grid services, renewable-energy coordination, battery degradation, communication performance, cybersecurity, interoperability, commercial viability, user participation, and energy equity. Particular attention is given to the boundary conditions and validation maturity underlying reported technical and economic outcomes, ranging from simulation and laboratory investigations to HIL testing, pilot demonstrations, and field implementations. Accordingly, the review aims to identify the cross-layer requirements and supporting evidence necessary for smart charging and V2G to progress from demonstrated technical potential to dependable pilot, network-scale, and commercial deployment. To achieve this aim, the review has three objectives:

- To develop a cross-layer analytical framework connecting batteries, chargers, distribution networks, control systems, communication infrastructure, markets, regulation, and vehicle users.
- To critically evaluate reported technical and economic benefits against their boundary conditions, including network configuration, vehicle availability, battery degradation, communication dependence, mobility requirements, baseline selection, and validation maturity.
- To propose an evidence-gated deployment pathway that distinguishes installed, connected, controllable, and contractually dependable fleet capability and defines the evidence required for progression from simulation to integrated pilots, network-scale operation, and system readiness.

1.3 Literature Search and Selection Criteria

This article adopts a critical integrative review rather than a systematic or meta-analytic design. The literature search was conducted using six scientific information sources: Scopus, Web of Science Core Collection, Institute of Electrical and Electronics Engineers (IEEE) Xplore, ScienceDirect, SpringerLink, and Google Scholar. The review primarily covered publications from January 2013 to August 2026, with the final literature search completed on August 30, 2026. The search strategy combined four principal concepts: EVs, smart or bidirectional charging, grid integration, and enabling control technologies. Relevant search terms were combined using Boolean operators (AND and OR), with the search syntax adapted to each information source. Peer-reviewed English-language journal articles were prioritized, while relevant peer-reviewed conference papers reporting technical developments or emerging concepts were also considered. Selected authoritative institutional reports and standards-related documents were retained where relevant to the technical, regulatory, or institutional context. Studies were selected based on their relevance to at least one layer of the interconnected battery–charger–grid–communication–market–user system, their technical or methodological contribution, and their relevance to the review objectives. Editorials, news articles, book reviews, patents, theses, opinion papers, and other non-peer-reviewed publications outside the specified institutional and standards-related exceptions were excluded. Duplicate records were identified by comparing article titles, author names, publication years, publication sources, and digital object identifiers, while backward reference searching was used to identify additional relevant studies. Each selected study was interpreted according to its system boundary, control objective, baseline, network and mobility assumptions, reported performance metrics, and validation environment. Because substantial methodological heterogeneity prevents meaningful statistical pooling, quantitative findings are presented as study-specific evidence rather than directly comparable performance estimates. The synthesis focuses on cross-layer dependencies, recurring trade-offs, and the evidence needed to progress from component-level feasibility to pilot, network-scale, and commercial deployment. The review was organized around six interconnected domains: battery performance, charging hardware, distribution networks, communication and control, market arrangements, and user participation. Studies were examined for their relevance to these domains and for the extent to which their findings could inform deployment decisions. Particular attention was given to the distinction between simulated performance, laboratory validation, and evidence from operational settings. The synthesis was interpretive rather than statistical, and the inclusion of a study did not imply that its findings were directly comparable with those of other studies.

To distinguish nominal fleet potential from deployable grid flexibility, this review assesses V2G capability through two connected dimensions: operational qualification and evidence maturity. The resulting analytical framework is presented in Figure 1.

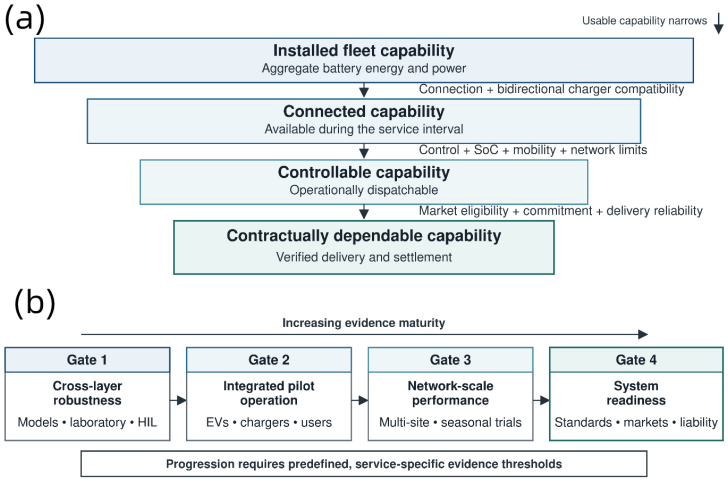


Figure 1. Evidence-gated framework for dependable V2G deployment: (a) capability qualification and (b) evidence progression

Note: V2G = vehicle-to-grid; SoC = state of charge; HIL = hardware-in-the-loop; EV = electric vehicle.

In Figure 1, Panel A illustrates how the aggregate installed battery capability of an EV fleet is progressively qualified into connected, controllable, and contractually dependable service capability. This qualification depends on physical connection, bidirectional charger compatibility, control authority, battery state-of-charge limits, mobility requirements, network and communication constraints, market eligibility, participation reliability, and verified service delivery. Panel B presents the corresponding progression of evidence from reproducible modeling, laboratory testing, and HIL validation to integrated pilots, network-scale field demonstrations, and system readiness. Available V2G

capability depends on the intended service, location, and time and should therefore be characterized by power, energy, duration, and response time. The widths shown in Panel A are conceptual and do not represent universal quantitative reductions.

For example, consider an EV fleet participating in a V2G program. Although the aggregate installed battery capacity represents the fleet's nominal energy potential, only vehicles connected during the relevant service period can contribute to grid support. The available capacity is further constrained by battery state-of-charge limits, minimum energy requirements for subsequent journeys, and bidirectional charger compatibility. Even among technically available vehicles, some owners may not authorize battery discharge or may withdraw from the program. Consequently, the energy that can be contractually committed for grid services represents only a proportion of the fleet's installed capacity. This example illustrates how the four stages of operational qualification in Figure 1 progressively distinguish nominal installed battery capacity from the energy flexibility that is connected, controllable, and contractually dependable for reliable service delivery. This qualitative illustration demonstrates the framework's logic rather than suggesting a universal quantitative reduction in fleet capacity.

2 Smart-Charging Technologies and Coordination Trade-Offs

The transition from passive, uncontrolled EV charging to coordinated smart charging is reshaping the relationship between electric transportation and power distribution networks. Smart charging serves as a cyber-physical coordination interface through which charging times and power levels can be adjusted according to grid conditions, electricity prices, renewable-energy availability, and user requirements [30]. It should therefore not be regarded merely as an improvement in charger technology. Its effectiveness depends on the combined performance of charging hardware, control strategies, communication infrastructure, distribution-network conditions, and user behavior. A strategy that improves peak-load management or charging cost may still require extensive data exchange, increase computational complexity, restrict user flexibility, or depend on assumptions that have not been validated at scale. Accordingly, smart-charging solutions should be evaluated not only by reported technical performance but also by scalability, communication and data requirements, compatibility with mobility needs, and the level of experimental validation.

2.1 Evolution From Uncontrolled to Smart Charging

Early EV integration studies commonly represented vehicles as passive grid-to-vehicle (G2V) loads that began charging immediately upon connection, without considering network conditions or system-level coordination. In one evaluated system, EV charging increased the total grid load by an average of approximately 18% [31]. This result illustrates the potential impact of additional charging demand under the study's specific assumptions rather than establishing a universally applicable value. More broadly, uncoordinated charging has been associated with higher demand peaks, increased power losses, voltage violations, voltage unbalance, harmonic distortion, transformer loading, and a possible reduction in transformer service life [32]. The severity of these effects nevertheless depends on EV penetration, charger power, charging schedules, spatial concentration, network topology, and existing feeder capacity.

These limitations motivated the development of coordinated smart-charging systems in which charging time and power are adjusted according to network conditions, electricity prices, infrastructure availability, and operational requirements [33]. More recent approaches apply deep RL and multi-agent coordination to the joint operation of PV generation, energy storage, and EV charging while accounting for interactions between transportation and power networks [34].

Nevertheless, increasing algorithmic complexity does not automatically establish practical superiority. Conventional rule-based or optimization-based strategies may offer greater transparency and lower communication and computational requirements, whereas learning-based controllers may provide greater adaptability under uncertain and time-varying conditions. Their performance, however, depends on training data, reward formulation, communication reliability, computational resources, safety constraints, and how well the training environment represents actual operation. The evolution from uncontrolled charging to artificial intelligence (AI)-assisted coordination should therefore be evaluated using consistent baselines, realistic mobility and network conditions, and progressively stronger validation, rather than isolated simulation results.

2.2 Charging Architecture, Power Levels, and Infrastructure Trade-Offs

Charging-infrastructure planning must jointly consider projected EV demand, station siting, network-expansion requirements, and voltage-dependent load behavior [35, 36]. At the equipment level, conductive charging is commonly grouped into Level 1 alternating current (AC), Level 2 AC, and direct current (DC) fast charging, although the terminology and power boundaries vary among standards and regions. Level 1 and Level 2 systems supply AC power through the vehicle's onboard charger and are generally suited to residential and workplace applications with relatively long parking durations. In contrast, DC fast charging bypasses the onboard charger and

supplies DC power directly to the battery, with reviewed implementations ranging from approximately 15 kW to 350 kW [37]. Although higher charging power reduces charging time, the concentration of high-power charging demand can increase transformer loading and thermal stress, particularly when EV charging coincides with existing residential demand [38].

This comparison highlights a fundamental infrastructure trade-off. Lower-power AC charging limits the instantaneous network burden but requires longer connection periods, whereas DC fast charging improves user convenience while increasing local capacity, equipment, and grid-connection requirements. Charger power should therefore be selected according to parking duration, feeder capacity, station location, expected utilization, and the intended charging service. Increasing charging power without considering these conditions may transfer the benefit of shorter charging time into greater network-reinforcement requirements or underutilized charging assets.

Charging infrastructure and operational coordination should be considered together because smart charging effectiveness depends on compatibility among charging equipment, parking duration, network capacity, and control requirements. While this section focuses on the infrastructure and coordination requirements of managed charging, bidirectional charger architecture and the control arrangements required for V2G operation are examined separately in Section 3.

2.3 Smart Charging Control and Coordination Strategies

Smart-charging strategies regulate the timing and power of EV charging to reshape the aggregate demand profile, typically pursuing objectives such as peak shaving, valley filling, and charging-cost reduction. Time-of-use (TOU) tariffs provide a simple and predictable mechanism for encouraging charging during predefined off-peak periods, whereas real-time pricing can respond more directly to changing electricity-system conditions [39]. However, price responsiveness should not be treated as equivalent to grid compatibility. The effectiveness of either mechanism depends on tariff design, vehicle availability, user response, aggregate charging demand, and local network capacity.

The spatial distribution of charging demand is also important. Charging-station siting is influenced by land use, transport activity, geographical context, and the availability of power infrastructure [40]. Consequently, a pricing strategy that improves the system-wide demand profile may still create local constraints when charging facilities are concentrated on particular feeders. Smart charging performance should therefore be evaluated at both the aggregate system and local network levels rather than solely by electricity cost reduction.

2.3.1 Price-based and incentive-based charging

Price-based charging introduces a fundamental coordination problem. Although lower prices can shift EV demand away from an existing peak, exposing many users or aggregators to the same price signal may synchronize their charging within the same low-price interval. This response can produce an “avalanche effect,” creating a new demand peak rather than eliminating the original concentration of load [41]. Price-based demand response is therefore not necessarily optimal from a grid perspective, even when it reduces individual charging costs.

The risk arises particularly when a uniform price signal is insufficiently updated or is applied without considering the resulting aggregate demand. Evidence indicates that dynamically updated demand-response signals can reduce the avalanche effect by adapting the charging incentive to changing system conditions [41]. Effective price-based coordination should therefore combine economic signals with aggregate-demand feedback, feeder constraints, vehicle availability, charging requirements, and users’ mobility needs. Its performance should be assessed not only by whether charging costs decrease or demand shifts, but also by whether the resulting load profile avoids new network constraints.

2.3.2 Optimization-based planning and scheduling

Optimization supports EV charging decisions across different time scales; these applications should therefore be distinguished when comparing their performance. At the infrastructure-planning level, a hybrid genetic algorithm–simulated annealing (GA–SA) method has been applied to determine charging-station locations in a PV-supported distribution network while reducing power losses and maintaining acceptable voltage levels [42]. Metaheuristic methods are useful for such multi-objective, non-linear search problems, but reported performance depends on parameter selection, network configuration, input data, comparison algorithms, and stopping criteria. Moreover, charging-station placement is a long-term planning problem rather than an operational scheduling problem; consequently, its results cannot be directly compared with those of real-time charging controllers.

At the operational level, mathematical optimization determines when and at what power EVs should charge to satisfy user requirements while improving the aggregate demand profile. The load-curve optimization reported by Majed et al. [39], for example, incorporates TOU and real-time electricity prices together with travel distance, driving periods, initial state of charge (SoC), and vehicle connection status. Such formulations allow charging schedules to be optimized under explicit mobility and operational constraints. Their effectiveness, however, depends on the accuracy of assumed electricity prices, mobility patterns, vehicle availability, and network conditions. Bidirectional scheduling and the provision of grid-support services through V2G are examined separately in Section 3.

Learning-based methods provide an alternative when charging decisions must adapt to uncertain or time-varying conditions. The adversarial reinforcement learning (ARL)–DGWO framework integrates adversarial RL with DGWO to coordinate real-time EV charging and energy trading. Under the investigated conditions, it reported an approximately 21% reduction in peak demand and an 18% improvement in renewable-energy utilization relative to the selected conventional approaches [18]. These results demonstrate the potential of adaptive coordination, but they remain conditional on the simulation environment, baseline methods, reward formulation, training data, and operating assumptions used in the study.

These approaches are complementary rather than directly interchangeable. Infrastructure-planning algorithms determine where to deploy charging capacity, deterministic scheduling methods identify operational decisions under explicitly formulated constraints, and learning-based controllers adapt decisions using observed system states. Direct performance comparison remains difficult because the studies address different decision time scales and employ different objectives, networks, datasets, constraints, and baselines. Reported numerical improvements should consequently be interpreted as scenario-specific evidence rather than proof that one algorithmic family is universally superior.

2.4 Communication and Computing Infrastructure for Smart Charging

Smart charging is a cyber-physical coordination problem requiring reliable information exchange among EVs, charging facilities, aggregators, computing platforms, electricity markets, and power-system operators. The exchanged information may include vehicle availability, charging requirements, network constraints, electricity prices, and charging schedules. A cloud–edge coordination framework for aggregated EV charging demonstrated that distributing computation across edge nodes can support dynamic charging optimization while reducing dependence on cloud-level data exchange [43]. The communication standards and interoperability requirements enabling these exchanges are examined in Section 4.1. However, compliance with individual standards does not necessarily guarantee end-to-end interoperability, as practical operation also depends on consistent implementation, compatible data models, reliable connectivity, and coordination across devices and platforms.

Cloud–fog–edge architectures distribute computing functions across multiple processing layers. Cloud platforms can provide fleet-wide visibility, long-term optimization, data storage, and market coordination, whereas edge and fog resources can process local information closer to EVs and charging facilities. In the cloud–edge EV aggregation case study, the proposed architecture reported a 61.7% improvement in its computational performance metric and a 98.6% reduction in cloud–edge information transfer relative to the study's baseline [43]. These results indicate potential scalability and communication benefits, but they remain dependent on the system architecture, fleet configuration, optimization problem, and evaluation conditions used in that study.

The value of layered computing also depends on the reliability of the complete communication path. A failure at the device, edge, fog, network, or cloud layer may interrupt the service even when the remaining components operate correctly; consequently, end-to-end availability must be evaluated across the complete multi-tier architecture [44]. One Internet-of-Things-enabled virtual power plant (VPP) framework reported response times below 50 ms using hybrid edge–fog computing [45]. Although this result demonstrates the potential of local processing to reduce latency, it should not be interpreted as a universal performance guarantee for EV charging systems because the reported response time was obtained under a particular architecture, resource configuration, and validation environment.

Greater connectivity also expands the cybersecurity exposure of EV charging infrastructure. Existing charging protocols and deployments may leave charging stations and EVs vulnerable, motivating the investigation of blockchain and AI for secure and decentralized charging coordination [46]. Nevertheless, blockchain should not be treated as a complete cybersecurity solution. Securing selected transactions or coordination functions does not by itself demonstrate that the integrated EV–grid system can maintain operation during communication disruption, component failure, or cyberattack. Communication latency, service availability, cybersecurity, interoperability, and recovery capability should therefore be evaluated as operational requirements of smart charging rather than as independent supporting functions.

2.5 Grid-Responsive and Renewable-Aware Charging

Renewable-aware charging coordinates EV demand with the temporal availability of solar and wind generation while respecting vehicle requirements and distribution-network constraints. This coordination involves more than connecting charging infrastructure to a renewable-energy source: it requires charging decisions to respond to changes in renewable output, aggregate demand, vehicle availability, and network capacity.

At the scheduling level, the Smart Renewable Optimization with Vehicle-aware RL (SRO–VRL) framework uses vehicle-aware, model-based RL to allocate charging power according to predicted renewable supply, EV SoC, and grid constraints. Relative to the benchmark reinforcement-learning and rule-based strategies considered in the study, the framework improved renewable-energy utilization by approximately 4%–5% and reduced peak grid demand

by approximately 7%–8% [47]. These results demonstrate the potential of adaptive renewable-aware scheduling, but they remain conditional on the investigated community-scale system, renewable-generation profile, benchmark methods, prediction accuracy, and assumed availability of charging data.

Grid-responsive charging can also incorporate power-quality objectives. In a MATLAB/Simulink study of a 10-MVA distribution feeder containing 20 EV chargers, an AI-based charging framework combining demand forecasting, reinforcement-learning control, and power-quality compensation reduced average total harmonic distortion (THD) from 6.8% to 4.6%, a decrease of 2.2 percentage points, and shortened the voltage-violation duration from 155 to 42 min [48]. Although these results indicate that charging coordination can support power-quality management, the improvements were obtained using a particular feeder model, charger population, baseline, controller configuration, and compensation equipment. These findings therefore demonstrate technical potential under simulation conditions rather than field-validated performance.

The reviewed studies address different dimensions of renewable-aware smart charging: Alshahr et al. [47] examine adaptive charging allocation, whereas Sabareeshwaran and Muniraj [48] evaluate charging coordination with power-quality compensation. Their reported outcomes should not be directly compared because they use different objectives, system boundaries, datasets, baselines, and validation environments.

Smart-charging approaches operate across hardware, infrastructure planning, operational control, computing, and grid support layers. Because the cited studies use different objectives, networks, baselines, and validation settings, their reported results cannot be compared directly. Table 1 synthesizes their technical functions, quantitative evidence, principal trade-offs, and remaining deployment gaps.

Table 1. Critical comparison of smart-charging approaches and deployment trade-offs

Approach	Function	Reported Evidence	Critical Trade-Off/Deployment Gap	Reference
Charging hardware	Set charging power; enable G2V/V2G	Uncontrolled demand: approximately +18% grid load; DC charging: approximately 15–350 kW	Charging speed and integration versus transformer and feeder stress	[31, 37–49]
Coordination architecture	Coordinate fleets centrally or locally	Centralized: system-wide information; decentralized: greater autonomy and scalability	Global coordination versus communication, computation, and user flexibility	[50]
Price-based scheduling	Shift demand through TOU and real-time pricing signals	Load shifting is achievable, but common price signals may create an avalanche effect	Cost reduction does not ensure network feasibility; feeder feedback is required	[39, 41]
Infrastructure planning	Locate and allocate charging stations	GA–SA reduced losses and maintained acceptable voltages in a PV-supported IEEE 33-bus system	Results are location-, model-, and parameter-specific; planning does not validate operation	[40, 42]
Learning-based control	Adapt charging under uncertainty	ARL–DGWO: 21% peak reduction and 18% higher renewable utilization	Training, reward, and baseline dependence; field transferability remains uncertain	[18]
Cloud–fog–edge control	Distribute computation and secure transactions	EV case study: 61.7% improvement in solving efficiency and 98.6% reduction in cloud–edge information transfer; separate VPP framework: sub-50-ms response	Efficiency versus availability, interoperability, and cybersecurity; evidence is architecture-specific	[43–46]
Renewable- and grid-aware control	Coordinate renewable use, PV extraction, and power quality	SRO-VRL: 7%–8% peak reduction and 4%–5% higher renewable utilization; feeder simulation: THD 6.8%→4.6% and voltage violations 155→42 min	Performance gains require forecasting, compensation, and added control; evidence remains simulation-specific	[47, 48]

Note: Reported values are study-specific and are not directly comparable because the cited studies use different objectives, baselines, system models, and validation conditions. G2V = grid-to-vehicle; V2G = vehicle-to-grid; DC = direct current; TOU = time-of-use; GA–SA = genetic algorithm–simulated annealing; PV = photovoltaic; IEEE = Institute of Electrical and Electronics Engineers; EV = electric vehicle; VPP = virtual power plant; SRO-VRL = Smart Renewable Optimization with Vehicle-aware Reinforcement Learning; THD = total harmonic distortion.

Table 1 shows that no individual approach establishes smart-charging deployment readiness. Improvements in

charging speed, load management, renewable utilization, or computational performance may introduce additional requirements for network capacity, communication reliability, interoperability, cybersecurity, and user flexibility. Deployment should therefore be evaluated through cross-layer system performance rather than isolated technological gains. This perspective provides the basis for examining V2G architecture, services, and operational trade-offs in Section 3.

3 Vehicle-to-Grid Integration: Architectures, Services, and Operational Trade-Offs

V2G integration extends managed charging by enabling bidirectional power exchange between EV batteries and the electricity grid [28]. With this capability, EVs can affect distribution network loading, voltage conditions, power quality, and operational flexibility [51]. However, bidirectional capability alone does not establish a dependable grid resource. Practical V2G performance depends on the coordinated operation of the battery, power-electronic interface, charging equipment, communication system, aggregator, grid operator, market mechanisms, and vehicle users. The central question is therefore not only whether an EV can return energy to the grid, but whether aggregated EV capacity can provide verifiable services without compromising mobility, battery health, network security, or commercial viability.

3.1 Evolution From Managed Unidirectional Charging to Bidirectional Vehicle-to-Grid

The role of EVs in power systems has evolved from uncontrolled charging to managed unidirectional charging (V1G) and, subsequently, bidirectional grid participation. Uncontrolled charging treats EVs as passive loads, whereas V1G adjusts charging time or power according to grid or user requirements. V2G extends this capability by enabling EVs to discharge energy and participate in grid-support services [52, 53].

Private vehicles remain parked for approximately 90%–95% of the day, suggesting substantial temporal flexibility [27]. Nevertheless, parking duration should not be equated with dispatchable V2G capacity. A parked vehicle may not be connected to a compatible bidirectional charger, may have an insufficient SoC, or may need to retain sufficient energy for an upcoming journey. Estimates based primarily on vehicle parking time may therefore overstate the capacity that an aggregator can reliably commit.

Early V2G research focused strongly on converter topology and the technical feasibility of bidirectional power exchange [54]. The deployment challenge has subsequently expanded to include fleet coordination, grid-service verification, market participation, communication reliability, battery degradation, and user acceptance. This progression demonstrates that V2G has evolved from a component-level power-conversion problem into a system-level coordination challenge. Its value should therefore be assessed in terms of dependable service capacity rather than installed battery capacity alone.

3.2 Vehicle-to-Grid Architecture and Operational Framework

A V2G system comprises several interdependent layers, including the EV battery, bidirectional converter, charging equipment, communication network, aggregator, market platform, and grid operator [29]. Failure or incompatibility at any layer can prevent service delivery, even when the remaining components function effectively. For example, an efficient converter cannot provide dependable grid support if communication is unavailable, the vehicle departs unexpectedly, or market participation is not permitted.

At the hardware layer, bidirectional onboard chargers provide an interface for controlled energy exchange between the vehicle and the grid [55]. At the market layer, research on interconnected microgrids demonstrates how incentive design can shape distributed energy trading [56]. Although this evidence is not exclusively V2G-specific, it illustrates that power-exchange capability does not, by itself, ensure economically coordinated participation. At the control layer, artificial neural network-based approaches have been proposed to support the intelligent integration of V2G and vehicle-for-grid functions [57]. These studies address different system layers and should therefore not be treated as directly comparable evidence of overall V2G readiness.

V2G operation is not governed by the power-conversion chain alone; it also depends on coordinated information, control, market, and user interactions. Figure 2 illustrates these cross-layer interfaces and the principal dependencies that determine whether fleet flexibility can be delivered reliably.

Figure 2 illustrates the physical layer, where DC power flows bidirectionally between the EV battery and the charger's DC-side interface, while the battery management system (BMS) enforces battery energy, power, and state-of-charge limits. The charger, or EV supply equipment (EVSE), interfaces with the distribution network through bidirectional AC power exchange. At the coordination layer, vehicle users communicate their mobility requirements, availability, and participation preferences, while the aggregator converts these inputs into charging and discharging schedules and receives operational telemetry. System operation and market-governance arrangements define network constraints, service requirements, prices, verification procedures, and settlement rules. Interoperability, cybersecurity, communication reliability, and measurement and verification are therefore cross-cutting requirements. Failure at any interface can prevent installed fleet capability from becoming a dependable V2G service.

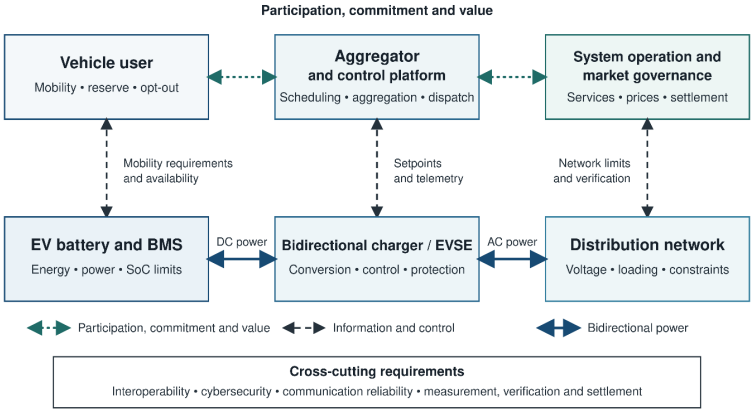


Figure 2. Cross-layer architecture for dependable V2G operation

Note: V2G = vehicle-to-grid; SoC = state of charge; DC = direct current; EVSE = electric vehicle supply equipment; AC = alternating current.

3.2.1 Bidirectional charger topologies

Bidirectional charger architectures provide the physical interface required for controlled power exchange between EV batteries and the grid. Integrated onboard charger–traction systems can reuse propulsion-system components during charging, potentially reducing component duplication and increasing power density while enabling bidirectional operation [49]. However, these benefits must be balanced against electrical-isolation requirements, control complexity, and compatibility between traction and charging functions. Segmented-winding machine-drive configurations have also been proposed to support high-power-density charging and torque-cancellation during V2G operation without mechanical vibration [58]. Dedicated converter configurations offer an alternative. A renewable-integrated isolated bidirectional converter has also been developed using an adaptive neuro-fuzzy inference system (ANFIS)-based controller to regulate battery charging and discharging during G2V and V2G operation. The configuration incorporates an incremental-conductance maximum power point tracking (MPPT) algorithm for PV power extraction and a dual-active-bridge converter for bidirectional energy transfer. DC-link voltage and load-power regulation were evaluated through simulation and HIL testing [59]. Dual-active-bridge converters have also been investigated for soft-switching bidirectional onboard charging under G2V and V2G operation [60]. Zeta-converter and synchronous-boost-based configurations have further been proposed to improve power quality, with the evaluated systems reporting THD values below 5% under their respective operating conditions [61, 62].

These findings demonstrate converter-level feasibility but should not be generalized directly to fleet-scale deployment. Performance obtained using a particular converter, controller, load condition, or laboratory arrangement does not establish equivalent power quality when numerous heterogeneous chargers operate on a constrained distribution feeder. Converter efficiency, thermal performance, protection coordination, grid-code compliance, and aggregate harmonic behavior must therefore be evaluated together.

3.2.2 Aggregation and operational coordination

Individual EVs have limited capacity and uncertain availability, restricting their ability to participate independently in electricity and ancillary-service markets. Aggregators address this limitation by coordinating distributed resources into a VPP and managing their participation in energy markets [63]. A sequential Stackelberg game has been used to develop personalized, network-aware charging prices for EV coordination [64].

Aggregation can improve market access and fleet-level coordination, but it also introduces forecasting, communication, settlement, and governance requirements. The aggregator must estimate vehicle availability, satisfy mobility constraints, account for battery degradation, meet committed service requirements, and distribute revenue among participating users. Consequently, the amount of battery capacity enrolled in a program does not adequately measure aggregation performance. More meaningful indicators include deliverable capacity, forecast error, service compliance, participant retention, communication availability, and net value after degradation and transaction costs.

3.3 Grid-Support Services and Availability Constraints

Aggregated EVs may provide frequency regulation, voltage support, peak shaving, valley filling, and related flexibility services [65, 66]. Priority-based V2G scheduling has also been investigated as a means of reducing power-grid load variance [67]. However, these services impose different requirements in terms of response time, power, energy, communication, and battery throughput. Evidence obtained for one service should therefore not be generalized to all V2G applications.

EV-aggregator support has been examined in multi-area frequency-stabilization frameworks containing renewable generation [68]. Model-predictive control has also been applied to EV aggregators participating in frequency-regulation markets [69]. This evidence does not directly demonstrate V2G performance; rather, it highlights the need to evaluate V2G alongside competing flexibility resources. Rapid EV response is valuable only when it is sufficiently reliable, available, and cost-effective compared with alternative technologies.

High-voltage direct-current (HVDC) systems offer another approach to frequency regulation in low-inertia grids [70]. Although this study does not evaluate V2G, it provides a useful reference when considering alternative sources of grid flexibility.

V2G chargers may also regulate active and reactive power to support distribution-network voltage [71, 72]. Bidirectional energy-flow control has been investigated to maintain acceptable operating conditions during EV charging and discharging [73]. Nevertheless, voltage-support performance depends strongly on charger location, feeder impedance, load distribution, EV penetration, available converter capacity, and local voltage constraints. Results obtained from one test feeder cannot establish equivalent performance on networks with different electrical characteristics.

Demand-response research in smart microgrids demonstrates the value of coordinating flexible loads and distributed generation [74]. Similarly, EV-focused studies identify peak shaving and valley filling as important V2G applications [75]. Decentralized coordination among EV aggregators, stationary storage, and distributed generation has also been shown to manage network losses and voltage conditions under the evaluated scenarios [76]. However, peak-support capability depends on whether vehicles are connected during the required service interval. Vehicle arrival and departure times, minimum state-of-charge requirements, travel demand, and owner participation may substantially reduce the capacity available during system peaks. Installed fleet capability should therefore be distinguished from its connected, technically controllable, and contractually dependable subsets.

3.4 Vehicle-to-Grid Energy-Management Architectures

V2G energy management can be organized through centralized, decentralized, or hybrid hierarchical architectures. These arrangements differ in their ability to coordinate fleet-wide charging and discharging decisions, manage communication requirements, accommodate local operating conditions, and preserve user flexibility. The broader literature on EV charging strategies also highlights the importance of evaluating technical, social, and regulatory feasibility when assessing coordination approaches [50]. Accordingly, no single architecture is universally preferable; its suitability depends on the intended grid service, fleet characteristics, information availability, and deployment environment.

3.4.1 Centralized control

Centralized control assigns charging and discharging decisions to a utility, system operator, or large aggregator. Access to fleet-wide information can support the coordinated optimization of operating costs, charging schedules, and network objectives [77, 78]. However, centralized operation requires extensive data collection, reliable connectivity, and sufficient computational capacity. It may also inadequately account for individual mobility preferences and privacy concerns.

Centralized control can provide effective system-level coordination when accurate and timely information is available. Nevertheless, claims of global optimality depend on model completeness, forecast accuracy, communication reliability, and the assumed availability of participating vehicles. A mathematically optimal schedule may remain operationally infeasible if users depart unexpectedly or communication is interrupted.

3.4.2 Decentralized and distributed control

Decentralized approaches allow EVs or charging stations to make decisions based on local measurements and limited coordination signals. Centralized training with decentralized execution has been applied to multi-agent V2G optimization under battery-degradation constraints [79]. Such structures may improve scalability and local autonomy while retaining a degree of system-level coordination during controller development.

However, decentralization does not eliminate the need to coordinate interactions among local controllers. Broader microgrid-control research shows that local control parameters and coordination mechanisms influence system stability and power sharing [80]. Although this evidence is not exclusively V2G-specific, it illustrates why independently optimized charging decisions may conflict when EVs operate under shared feeder constraints. Decentralized control must therefore incorporate mechanisms to prevent synchronized or competing actions from increasing power losses, voltage deviations, or network instability.

3.4.3 Hybrid hierarchical control

Hybrid hierarchical control combines higher-level system coordination with lower-level local decision-making. Hierarchical optimization has been used to coordinate EV charging across integrated medium- and low-voltage networks while accounting for user satisfaction [81]. This structure can allocate system-level objectives to regional aggregators or EV clusters while allowing local controllers to accommodate individual vehicle requirements.

However, additional control layers introduce interfaces through which delays, inconsistent objectives, or inaccurate information may affect system performance. Hybrid control should not therefore be assumed to combine only the advantages of centralized and decentralized systems. Its deployment value depends on clearly allocated responsibilities, compatible data exchange, robust coordination during communication failures, and validation under uncertain vehicle behavior.

All three architectures are affected by distribution-network capacity, interoperability, and behavioral uncertainty. High EV penetration can create congestion in networks originally designed for predominantly unidirectional power delivery [82]. Differences in communication standards and charging topologies complicate cross-platform operation [83], while uncertainty in vehicle arrival and departure times, charging demand, and travel-energy requirements limits the transferability of scheduling models [84]. Architecture selection should therefore consider network capacity, fleet scale, communication capabilities, interoperability, privacy requirements, and the level of uncertainty that the intended service can tolerate.

3.5 AI-Enabled Vehicle-to-Grid Optimization and Validation Limitations

Uncertainty in vehicle availability, charging demand, renewable generation, electricity prices, and battery condition has motivated the application of stochastic optimization and intelligent control. Scenario-based multi-objective mixed-integer linear programming has been applied to large V2G-enabled fleets while accounting for battery degradation and lifecycle emissions [85]. EV behavior modeling is also increasingly important because scheduling performance depends on whether simulated mobility patterns reflect actual users [86]. Multi-objective charging strategies have been used to balance distribution-network requirements and EV-user objectives [87]. Metaheuristic methods have also been applied to EV integration problems involving several competing objectives [88]. In a mean-case scenario examined in one study, the fitness-sorted moth search algorithm achieved an improvement of more than 81% over the selected particle-swarm and genetic-algorithm benchmarks in minimizing the specified combination of operational and environmental costs [89].

This numerical result should not be interpreted as evidence of universal algorithmic superiority. Its magnitude depends on the objective formulation, component weighting, parameter settings, test network, comparison methods, and stopping criteria. More generally, mathematical programming provides explicit constraint representation, whereas learning-based and metaheuristic methods can address adaptive or nonlinear decision problems. Their reported outcomes are not directly comparable when studies use different networks, datasets, constraints, objectives, and baselines.

For deployment-oriented evaluation, algorithmic performance should be assessed beyond the environment in which the method was developed. Relevant evidence should include sensitivity to forecast errors, previously unseen mobility patterns, communication delays, device failures, network-model mismatches, computational burden, and safety constraints. Without such validation, improved simulation performance represents technological potential rather than demonstrated operational readiness.

3.6 Bidirectional Coordination With Renewable Energy

V2G can provide temporal flexibility in PV-dominant energy systems by shifting EV charging toward periods of renewable availability and, where permitted, discharging during periods of lower generation or higher demand [90]. However, the availability of this flexibility depends on the temporal alignment of renewable-energy output, vehicle connection, battery SoC, and grid demand.

The practical contribution of V2G also depends on users' willingness to make their batteries available. Moreover, the factors influencing V2G acceptance have evolved alongside the technology and its proposed applications [91]. Renewable-integration potential should therefore not be estimated solely on the basis of technical battery capacity.

Scenario-based analysis incorporating heterogeneous driver profiles has indicated that tailored V2G programs could reduce peak net load by up to 22.9% by 2030 under the behavioral and technical assumptions evaluated [92]. Year-long residential microgrid simulations have also reported reductions of up to 5.7% in the levelized cost of electricity, with greater benefits as the share of solar generation increased [93].

These findings demonstrate system-level potential but remain scenario-specific. The reported benefits depend on renewable generation profiles, vehicle availability, user participation, battery and charger capacities, network constraints, price assumptions, and the selected baseline. V2G should therefore be treated as one component of renewable-energy coordination rather than as an independent guarantee of high renewable penetration.

V2G applications are frequently presented as offering a common set of grid benefits, although the conditions under which they are delivered differ substantially. Table 2 compares the principal applications by operating function, implementation requirements, and constraints on dependable service provision.

Table 2 demonstrates that V2G service capability cannot be inferred solely from installed battery capacity. Each application requires a different combination of vehicle availability, converter capacity, communication performance, operational coordination, and user participation. Deployment assessments should therefore distinguish installed

fleet capability from its connected, controllable, and contractually dependable subsets. Battery degradation further influences these constraints, as discussed in Section 3.7.

Table 2. Critical comparison of V2G applications and dependable-delivery constraints

Application	V2G Function	Principal Delivery Requirements	Critical Constraint	Reference
Aggregator-mediated market participation	Coordinate distributed resources through a VPP and manage pricing and scheduling for EV clusters	Availability forecasting, reliable communication, settlement, and participant compensation	Enrolled battery capacity may exceed market-deliverable capacity	[63, 64]
Frequency regulation	Adjust aggregated charging or discharging in response to frequency requirements	Connected vehicles, state-of-charge margin, low-latency control, and aggregator coordination	Rapid device-level response does not necessarily guarantee dependable fleet-level capacity	[68, 69]
Voltage support	Regulate active and reactive power in distribution networks	Suitable charger location, available converter capacity, and feeder information	Performance depends strongly on feeder characteristics and local operating conditions	[71–73]
Peak shaving and load-variance reduction	Coordinate charging and discharging to reduce peak demand and load variation	Vehicle availability, predictable mobility, and minimum state-of-charge requirements	Mobility requirements and user participation constraints reduce dependable peak-support capacity	[67, 75]
Renewable-energy coordination	Align charging and discharging with variable renewable generation	Temporal coincidence among renewable output, grid need, vehicle connection, and user participation	Reported benefits remain dependent on scenario assumptions, user behavior, and system characteristics	[90–93]

Note: The cited studies examine different V2G applications, system boundaries, and operating conditions. The identified requirements and constraints should therefore be interpreted within their respective contexts. VPP = virtual power plant; EV = electric vehicle; V2G = vehicle-to-grid.

3.7 Battery Degradation and Battery-Aware Operational Trade-Offs

Battery degradation directly affects the value proposition of V2G participation for EV owners. The interaction between V2G operation, grid-service provision, and battery performance is therefore an important consideration when assessing the overall benefits and limitations of bidirectional EV integration [65]. User-centric V2G coordination has been investigated as a means of balancing grid benefits against battery-degradation costs [94]. Studies examining different driving scenarios, peak-shaving requirements, and frequency-regulation services demonstrate that degradation outcomes depend on battery operating conditions rather than on V2G participation alone [95].

Experimental evidence further shows that V2G-related battery degradation is influenced by battery chemistry, temperature, state-of-charge range, depth of discharge, cycling frequency, and charging and discharging rates [96]. A single degradation value should therefore not be applied across different battery types, grid services, or operating strategies.

Battery degradation also directly affects vehicle owners' willingness to participate in V2G programs. Owners must consider whether the expected financial compensation adequately offsets the incremental battery wear associated with providing grid services. Frequent cycling and greater depths of discharge may increase battery-degradation costs, although the extent of this effect depends on operating conditions [95, 96]. Because EV owners must balance the financial benefits of V2G participation against potential battery deterioration, participation may become less attractive when expected returns do not adequately compensate for incremental degradation [94], [97]. Accordingly, compensation mechanisms should account for service-specific energy throughput and estimated incremental degradation rather than rely solely on fixed participation payments. Transparent and well-structured compensation mechanisms can therefore improve the economic attractiveness of participation while supporting a more equitable distribution of V2G benefits and costs.

Battery-aware control involves an unavoidable trade-off between service provision and battery preservation. Increasing battery cycling availability can expand grid-support capability but may also accelerate battery wear, whereas restrictive operating limits can protect battery health while reducing service capacity and V2G revenue. Scheduling decisions should therefore account for battery condition, degradation costs, mobility demand, service

requirements, and compensation. A V2G program cannot be considered economically sustainable if grid benefits are achieved by transferring the risk of battery degradation to vehicle owners without adequate compensation.

3.8 Vehicle-to-Grid Communication, Cybersecurity, and Performance Trade-Offs

Communication and cybersecurity affect both operational reliability and public confidence in V2G systems. The broader social dimensions of V2G deployment indicate that trust, responsibility, and user acceptance are closely connected to the system's technical design [98]. Secure information exchange among EVs, chargers, aggregators, and grid operators is essential because charging schedules and grid-service commitments depend on accurate and timely data.

International Organization for Standardization 15118-based smart-charging implementations demonstrate structured communication between EVs and charging infrastructure [99]. Nevertheless, the existence of a communication standard does not, by itself, guarantee end-to-end interoperability. Recent reviews continue to identify implementation differences, cybersecurity concerns, and unresolved integration requirements across the V2G ecosystem [100].

Internet of Vehicles (IoV) connectivity also exposes V2G systems to cyberattacks that may compromise confidentiality, integrity, and service availability. Intrusion-detection research has consequently examined advanced AI methods for identifying threats in IoV environments [101]. However, detection performance on a selected dataset does not necessarily demonstrate resilience to coordinated attacks or communication outages.

Blockchain-based anonymous reward mechanisms have been proposed to support privacy-preserving incentives in V2G networks [102], while lightweight blockchain frameworks have been developed to facilitate data sharing and energy trading [103]. These approaches may improve transaction integrity, traceability, and privacy; however, they should not be regarded as comprehensive cybersecurity solutions. Their suitability also depends on latency, computational requirements, scalability, governance, interoperability, and recovery capabilities.

Accordingly, V2G cybersecurity should be evaluated across the entire EV–charger–aggregator–market–grid chain. Deployment readiness requires not only secure transactions but also safe and continuous operation, fault detection, fallback control, and service recovery when communication or computing components become unavailable. The evidence reviewed in this section demonstrates that V2G readiness depends on integrated performance across power conversion, fleet availability, control, communication, battery health, market coordination, and user participation rather than on the isolated performance of individual technologies.

Because the reviewed studies use different system boundaries and validation settings, their reported outcomes require cautious interpretation. Table 3 summarizes selected findings and the principal limitations that prevent direct inference of deployment readiness.

Table 3. Critical interpretation of selected V2G findings and their deployment limitations

Domain	Reported Evidence	Critical Interpretation	Reference
Power quality	THD below 5% in the evaluated converter configurations	Demonstrates component feasibility; fleet-scale power quality remains unverified	[61, 62]
Optimization	The fitness-sorted moth search algorithm achieved an improvement of more than 81% over the selected particle swarm optimization and GA benchmarks	Results depend on the objective, baseline, parameters, and test system	[89]
Renewable flexibility	Peak net load reduced by up to 22.9% by 2030	Benefit remains dependent on behavioral and technical assumptions	[92]
Microgrid economics	Levelized electricity cost reduced by up to 5.7%, with greater benefits at higher solar shares	Profitability after accounting for infrastructure and battery-degradation costs has not been established	[93]
Battery degradation	Degradation varies with service, chemistry, temperature, SoC, and cycling conditions	A universal battery-degradation factor cannot be applied across different V2G programs	[94–96]
Communication and cybersecurity	ISO-based communication, IoV intrusion detection, and blockchain transactions have been investigated	Interoperability, cybersecurity resilience, and large-scale service recovery remain insufficiently validated	[99–103]

Note: The reported values are study-specific and are not directly comparable because the cited studies use different objectives, baselines, system models, and validation conditions. THD = total harmonic distortion; GA = genetic algorithm; SoC = state of charge; ISO = International Organization for Standardization; IoV = Internet of Vehicles; V2G = vehicle-to-grid.

Table 3 indicates that current evidence provides stronger support for the technical feasibility of individual V2G functions than for integrated deployment readiness. Field-scale validation across hardware, communication, grid operation, battery health, markets, and user participation is therefore necessary. These cross-layer requirements are examined further in Section 4, with particular attention to interoperability, regulation, market participation, and user acceptance.

4 Deployment Conditions: Interoperability, Policy, Markets, and Participation

Large-scale V2G deployment requires more than effective chargers and control algorithms. It depends on a sociotechnical framework that connects standards, communication protocols, market regulation, infrastructure investment, business models, and consumer participation [104, 105]. Although enabling technologies and potential system benefits are increasingly documented, unresolved institutional and implementation issues continue to impede the transition from technical feasibility to scalable operation [106]. Moreover, V2G flexibility is participation-dependent: installed battery capacity has limited operational value when vehicles are unavailable, or users are unwilling to permit charging and discharging [107]. Deployment readiness should therefore be evaluated through the coordinated performance of technical, regulatory, commercial, and social systems.

4.1 Interoperability Standards and Implementation Gaps

V2G requires coordinated information exchange across interfaces involving vehicles, chargers, management platforms, aggregators, and grid operators. Existing standards and intelligent energy-system architectures provide important foundations for this exchange [108]. However, compliance with a single protocol does not establish interoperability across the entire communication chain.

The Open Charge Point Protocol (OCPP) facilitates communication between charging stations and central management platforms, reducing dependence on proprietary charger-management interfaces [107]. International Electrotechnical Commission (IEC) 61850 provides a complementary framework for modeling EV charge–discharge management within smart-grid and distributed-energy-resource environments [109]. These protocols address different interfaces and functions; consequently, implementing one does not eliminate the need for compatibility with the others.

Regional differences in charging infrastructure, connector systems, power ratings, and supporting energy resources create an additional harmonization challenge [110]. A configuration developed for one charging ecosystem may require hardware, software, certification, or grid-code modifications before it can be transferred to another market. Interoperability should therefore be demonstrated through end-to-end testing across vehicles, chargers, management platforms, aggregators, and grid operators rather than inferred from nominal compliance with individual standards.

Interoperability also has a user-facing dimension. Evidence from Taiwan indicates that V2G acceptance is influenced by expected performance, perceived value, battery-related concerns, and the degree of control retained by users [111]. Although this evidence concerns user behavior rather than protocol-level interoperability, it shows that technically interoperable systems must also deliver predictable and understandable services. A system that exchanges data successfully but provides unclear control arrangements or compensation cannot be considered fully deployment-ready.

Cybersecurity further complicates standardization. Man-in-the-middle attacks can compromise vehicular communications by modifying, delaying, or intercepting transmitted information [112]. Broader IoV research also identifies security and architectural challenges across connected vehicles and their supporting networks [113]. Stronger authentication may increase computational and communication requirements, whereas low-latency communication without adequate security measures may expose V2G operations to manipulation. Standardization must therefore balance interoperability, latency, cybersecurity, computational burden, and recovery capabilities rather than optimize any single requirement independently.

4.2 Regional Policy and Regulatory Landscape

Regional policy frameworks differ in their decarbonization objectives, electricity-market structures, infrastructure conditions, and stages of EV adoption. China's EV development pathway, for example, has been examined in relation to its carbon-neutrality target, provincial energy-system differences, emissions, and cost-effectiveness [114]. More recent evidence indicates rapid growth in China's new-energy vehicle market penetration and associated changes in transportation energy consumption [115]. These developments demonstrate how coordinated policies and infrastructure investments can accelerate EV adoption.

However, high EV penetration does not, by itself, establish a functional V2G market. Bidirectional deployment also requires interoperable chargers, market-access rules, clearly defined aggregator responsibilities, discharge-pricing mechanisms, and consumer participation. Adoption statistics should therefore not be interpreted as evidence of V2G readiness.

The United Kingdom has used financial incentives to support electric-vehicle adoption, although their effect depends on broader economic conditions [116]. Evidence from the European Union similarly indicates that reductions in purchase incentives can influence electric-vehicle adoption [117]. Although purchase incentives can accelerate market entry, they do not necessarily encourage sustained V2G participation because vehicle ownership and battery availability for grid services are separate decisions.

Distribution-network preparation is equally important. A California case study projected that nearly 20% of distribution feeders would require upgrades under the EV adoption conditions evaluated [118]. Although this result is specific to the region and scenario examined, it demonstrates that transportation policy cannot be separated from network planning. Policies that accelerate EV adoption without corresponding investments in distribution capacity may shift infrastructure-upgrade costs and operational risks onto utilities and electricity consumers.

V2G-specific policies must also coordinate facility scale, subsidies, and discharge pricing. Joint decision-making models show that infrastructure investment and pricing incentives interact in determining V2G participation and deployment outcomes [119]. Subsidies should therefore be evaluated not only by the number of EVs or chargers they support but also by whether they enable secure bidirectional operation, provide appropriate compensation for flexibility, and allocate infrastructure costs transparently.

Overall, regional experience does not support a universally transferable policy model. Effective policy frameworks must reflect local electricity markets, feeder capacity, charging access, mobility patterns, and regulatory institutions. The desired policy outcome is not simply rapid EV adoption but the creation of conditions under which V2G services can be delivered reliably and economically.

4.3 Consumer Acceptance, Mobility Concerns, and Participation Barriers

V2G depends on voluntary access to privately controlled batteries. Empirical research shows that willingness to participate is influenced by perceived benefits, risks, technical understanding, and the conditions associated with vehicle use [120]. Models that assume continuous or predictable EV availability may therefore overestimate practical flexibility if they do not account for user preferences, mobility requirements, and the ability to opt out.

Charging coverage and range anxiety remain important constraints. National infrastructure analyses identify geographic gaps in public charging availability [121], while research on long-distance travel demonstrates that charging behavior and range anxiety influence user decisions [122]. V2G may intensify these concerns if users believe that providing grid services could leave insufficient battery energy for unexpected journeys. Participation mechanisms should therefore include minimum state-of-charge guarantees, departure-time requirements, user override options, and transparent information about battery use.

Public understanding also varies substantially. Evidence from Nordic focus groups identifies differences in how consumers perceive EV and V2G benefits, risks, and responsibilities [123]. Latent-class analysis further demonstrates that EV users exhibit different charging patterns and behavioral characteristics [123], as well as different charging styles and behavioral characteristics [124]. These findings indicate that no single tariff, control setting, or communication strategy is likely to suit all participants. V2G programs may therefore require differentiated participation options rather than assuming a uniform consumer response.

Physical access can constrain participation regardless of willingness. Research from the Netherlands demonstrates the social and operational complexity of shared charging arrangements among neighbors and users without dedicated residential parking [125]. Studies of multifamily housing also identify practical challenges affecting the provision of residential charging infrastructure [126]. Users without private parking or dedicated chargers may therefore be excluded from V2G even when they are willing to participate.

Technical coordination models do not always adequately account for these behavioral constraints. Fully decentralized multi-agent RL has been applied to EV aggregation for frequency regulation [127]. Such work demonstrates the potential for algorithmic coordination but does not establish whether users will accept the resulting dispatch decisions, compensation arrangements, or battery use. Consumer behavior should therefore be incorporated into controller constraints, availability forecasts, market design, and program evaluation rather than treated as an external uncertainty after system development.

4.4 Aggregator Business Models, Compensation, and Value Distribution

Aggregators can combine the limited and uncertain capacity of individual EVs into a market-facing flexibility resource. Reviews of V2G participation in electricity markets identify potential opportunities but also highlight forecasting, communication, market-access, and regulatory challenges [128]. Aggregation lowers the participation barrier for individual owners while assigning aggregators responsibility for coordinating dispatch, meeting service commitments, and distributing revenues.

V2G coordination has also been investigated using game-theoretic strategies to provide ramping flexibility to power systems [129]. Such models can represent strategic interactions among participants, but their outcomes depend on assumed prices, information availability, user rationality, and market rules. A mathematically stable coordination

outcome does not necessarily establish commercial viability when actual participation is uncertain. System-level savings must therefore be distinguished from vehicle-owner returns because modeled reductions in total operating costs do not automatically translate into realizable income for participants [128, 129].

Battery degradation further separates gross system value from net participant benefits. General battery research identifies the growth of the solid electrolyte interphase (SEI) as an important mechanism affecting lithium-ion battery aging during cycling [130]. A strategy that increases battery throughput to maximize short-term system savings may therefore reduce long-term benefits for vehicle owners. Conversely, overly restrictive battery-preservation constraints may reduce grid flexibility.

Optimization research has explicitly considered compensation for battery degradation while maximizing V2G benefits [97]. This represents an important progression from system-cost minimization toward participant-aware value allocation. Nevertheless, commercial credibility requires transparent answers to four questions: Who provides flexibility? Who receives the revenue? Who bears the battery and infrastructure costs? Who is responsible when committed capacity is unavailable?

A viable business model should therefore evaluate the net value generated for aggregators, utilities, market operators, and vehicle owners. Total system savings alone are insufficient to establish commercial viability when compensation is inadequate, battery-degradation costs are transferred to users, or mobility requirements are not adequately protected.

4.5 Energy Equity and Social Inclusion

EV adoption is influenced by urbanization, household income, and socioeconomic inequality [131]. If V2G deployment follows similar patterns, its financial and resilience benefits may become concentrated among households that already own suitable vehicles, have private parking, and can install compatible bidirectional chargers. Evidence from the United States identifies income- and race-related disparities in access to publicly available charging infrastructure [132]. These findings demonstrate that the total number of chargers does not adequately reflect equitable access. Infrastructure may appear sufficient at the regional level while remaining physically, financially, or operationally inaccessible to certain communities.

Charging-station planning involves site selection, capacity sizing, network upgrades, and the management of multiple sources of uncertainty [133]. Equity should be incorporated into these planning decisions by examining which communities receive infrastructure, which users can participate in flexibility markets, and how network-upgrade costs are distributed. This distinction is important because a technically optimal location is not necessarily an equitable one.

Environmental equity also extends beyond access to charging infrastructure. Battery production and end-of-life management create resource-use and recycling challenges across the EV lifecycle [134]. V2G may improve renewable-energy utilization and operational flexibility, but these benefits should not be evaluated in isolation from the material and environmental burdens associated with battery production, extended use, and disposal.

Research examining EV and V2G co-benefits beyond emissions and economic considerations highlights broader social and system-level outcomes [135]. These findings indicate that equity should not be treated as a secondary consequence to be assessed after deployment. Instead, it should be incorporated as a design criterion alongside cost, reliability, interoperability, battery health, and user mobility.

Overall, standards, policies, markets, consumer behavior, and equity are strongly interconnected. Standards provide the basis for bidirectional operation, but fragmented implementation restricts system-level interoperability. Policies can accelerate EV adoption, but adoption incentives do not automatically establish V2G participation or market access. Aggregators may create revenue opportunities, but commercial viability depends on dependable vehicle availability, transparent compensation, and battery-aware value allocation. Finally, unequal access to charging infrastructure may prevent lower-income households, rural communities, and residents of multifamily housing from participating in or benefiting from V2G. These unresolved sociotechnical conditions underpin the research priorities and staged deployment pathways presented in Section 5.

The deployment barriers examined in Sections 4.1–4.5 are interdependent rather than isolated. Table 4 synthesizes the principal requirements and unresolved readiness gaps across five domains: interoperability and cybersecurity, policy and network preparation, consumer participation, aggregator viability, and energy equity. This system-level comparison demonstrates why technical feasibility alone is insufficient to establish readiness for scalable V2G deployment.

As summarized in Table 4, progress in one deployment domain can be constrained by deficiencies in another. Interoperable equipment cannot provide dependable grid services without secure communication, regulatory authorization, and market access. Similarly, attractive compensation cannot mobilize flexibility without user trust, mobility protection, and equitable charging access. Large-scale V2G deployment therefore requires coordinated technical, regulatory, commercial, and social validation rather than isolated optimization within individual domains. These interconnected requirements provide the basis for the research priorities and staged deployment pathways

discussed in Section 5.

Table 4. Critical synthesis of V2G deployment conditions and readiness gaps

Domain	Deployment Requirement	Principal Readiness Gap	Reference
Interoperability and cybersecurity	End-to-end compatibility, secure communication, and recovery capability	Protocol compliance does not guarantee system-level interoperability or resilience	[108–113]
Policy and grid preparation	Market-access rules, discharge pricing, infrastructure investment, and network planning	EV-adoption incentives do not automatically establish V2G readiness	[114–119]
Consumer participation	Mobility guarantees, opt-out control, understandable compensation, and charging access	Technical battery capacity may exceed practically available flexibility	[120–127]
Aggregator and market viability	Availability forecasting, service compliance, battery-aware compensation, and transparent value distribution	System-level savings may not provide positive net value to vehicle owners	[97, 128]
Equity and lifecycle sustainability	Inclusive infrastructure planning and consideration of battery-lifecycle burdens	V2G benefits may remain concentrated among users with vehicles, private parking, and compatible chargers	[131–135]

Note: The cited evidence is drawn from different regional, technical, market, and social contexts. The identified readiness gaps are qualitative and should not be interpreted as directly comparable measures. EV = electric vehicle; V2G = vehicle-to-grid.

5 Research Priorities and Deployment Pathways

5.1 Technical Validation and Cross-Layer Robustness

V2G deployment remains constrained by insufficient validation across interconnected infrastructure, battery, power-electronic, communication, and control layers. Research on charging-station planning addresses infrastructure siting, capacity sizing, network upgrades, and planning uncertainty [133], while battery-health-aware optimization highlights the need to balance grid benefits, degradation costs, and participant incentives [136]. However, reported outcomes remain sensitive to assumptions about charging demand, vehicle availability, battery condition, electricity prices, and user participation. Future studies should explicitly disclose these assumptions, quantify uncertainty, and report both system-level and vehicle-owner outcomes.

Power quality and cybersecurity also require system-level evaluation. Existing monitoring methods can support the detection, classification, and localization of power-quality disturbances [137]; however, high detection accuracy does not necessarily demonstrate effective mitigation. Similarly, Biswas et al. [138] specifically address man-in-the-middle attack detection and should not be interpreted as evidence of resilience against other types of cyber threats. Validation should therefore examine voltage deviations, harmonic distortion, communication latency, false-alarm rates, recovery time, and fail-safe operation under realistic network conditions involving multiple vendors.

Power HIL testing can reveal the operational consequences of time delays that offline simulations may not capture [139]. Nevertheless, it represents an intermediate level of evidence rather than a substitute for field validation. A credible validation pathway should progress from simulation and HIL testing to controlled pilots and multi-site demonstrations, with consistent reporting of grid-service delivery, communication performance, mobility violations, operational failures, and differences between predicted and observed outcomes.

5.2 Interoperability, Regulation, and Market Design

V2G interoperability extends beyond physical connector compatibility to include the communication and exchange of data and control commands among vehicles, chargers, aggregators, and grid operators. Existing research identifies gaps in protocol alignment, standardization, certification, and market adoption [140]. Razzaque et al. [141] further document cybersecurity vulnerabilities across interconnected V2G systems, while Ruttala et al. [142] demonstrate the diversity of charger topologies, standards, and commercial implementations. Future work should therefore prioritize multi-vendor conformance testing, protocol compatibility, secure authentication, and fallback operation during interface or communication failures.

Market design must align grid requirements with participant value. The economic viability of V2G is strongly influenced by charging-tariff structures [24]. Research on price-responsive EV charging also shows that conventional

demand-response signals may create rebound peaks or avalanche effects [41]. Related research proposes personalized, network-aware charging prices [64], whereas V2G-specific research examines the joint design of facility scale, discharge prices, and government subsidies [119]. These approaches should be evaluated in terms of participant revenue, network constraints, degradation compensation, and sensitivity to behavioral assumptions. Future regulatory research should also clarify market eligibility, data responsibilities, settlement procedures, and liability for service non-delivery.

5.3 Participation, Equity, and Distributional Evaluation

The operational flexibility available to an aggregator depends not only on installed battery capacity but also on vehicle owners' willingness to participate and vehicle availability. Evidence from reviews, surveys, and focus groups identifies guaranteed mobility, perceived risk and control, battery-degradation concerns, financial value, trust, and clear information as important conditions for V2G acceptance [91, 98, 111, 120, 123]. Future trials should therefore report participant recruitment and withdrawal, opt-out frequency, connection time, mobility overrides, compensation, and the difference between committed and delivered capacity rather than assuming that all enrolled vehicles remain continuously available.

Equitable participation also depends on access to charging infrastructure. Existing evidence identifies differences in home and workplace charging accessibility [8], geographic coverage [121], shared residential charging arrangements [125], and charging provision in multi-family housing [126]. Income inequality may also hinder EV adoption [131], while income- and race-related disparities affect access to public charging infrastructure [132]. Although these studies examine EV adoption and charging rather than V2G participation directly, they identify important prerequisites for inclusive V2G deployment. Because EV and V2G benefits extend beyond cost savings and emissions reductions [135], future evaluations should examine who can participate, who bears equipment and battery-degradation costs, and how financial, mobility, and resilience benefits are distributed.

5.4 Emerging Technologies: Readiness Versus Novelty

Battery digital twins can integrate physical models, operational data, and AI to support battery-state estimation and management [143, 144]. However, these studies demonstrate technical potential rather than field-scale V2G readiness. Future evaluations should examine calibration accuracy, estimation uncertainty, computational latency, transferability across battery chemistries, and performance improvements over simpler estimation methods.

Emerging V2G applications also include deep-learning-supported gravity storage in residential microgrids [145] and quantum-secured mutual authentication [146]. Although promising, these individual studies do not establish scalability, multi-vendor interoperability, or operational reliability. These approaches should therefore be evaluated through reproducible testing and realistic pilot deployments rather than relying solely on model performance.

Second-life battery storage and recycling contribute primarily to lifecycle management rather than real-time V2G operation [147, 148]. Similarly, multi-sensor fault detection and federated learning have been studied in broader smart-grid contexts [149, 150], but their specific value for V2G applications remains to be demonstrated. Future research should evaluate these technologies in terms of field performance, cybersecurity, lifecycle impacts, cost, and deployment readiness rather than novelty alone.

5.5 Gate-Based Roadmap for Scalable Vehicle-to-Grid Deployment

A scalable V2G roadmap should be based on evidence thresholds rather than fixed timelines. A broader systems perspective requires technical performance to be evaluated alongside its social, economic, and environmental implications [151]. The first gate should establish reproducible performance across battery, converter, communication, and cybersecurity systems under clearly reported operating conditions.

The second gate should validate integrated fleet and market operations, including aggregator control, service delivery, participant compensation, and alignment with user charging preferences [152, 153]. The third gate should demonstrate grid-support performance at a larger operational scale. Relevant evidence includes frequency support from EV charging stations [154], communication and scheduling architectures for multi-microgrid coordination [155], and reliable battery state-of-health estimation [156]. Because these studies address different system components, their findings should not be interpreted as evidence of complete V2G readiness without integrated testing.

The final gate should address coordinated transportation–power planning and overall system readiness. Existing reviews identify the need for stronger integration across transportation, power, and distributed energy systems [157, 158], while stakeholder evidence from the United States highlights continuing barriers involving standards, warranties, vehicle availability, and institutional support [159]. Deployment should proceed only when technical reliability, participant value, interoperability, clearly defined policy responsibilities, and field performance have been demonstrated at the intended scale.

The preceding discussion defines the evidentiary requirements of each deployment gate. Table 5 complements this analysis by translating the proposed framework into observable indicators and criteria for progression between stages.

Table 5. Evidence-gated deployment criteria and their application to V2G services

Deployment Gate	Key Verification Criteria	Service-Specific Examples	Advancement Decision
Cross-layer robustness	Battery degradation, conversion efficiency, power quality, communication latency and availability, and cybersecurity resilience.	Frequency regulation: communication latency and availability; peak shaving: conversion efficiency and power quality	Reproducible performance within service-specific limits
Integrated operation	Committed and delivered capacity, user participation, mobility compliance, and interoperability	Frequency regulation: delivered fleet capacity; renewable integration: vehicle availability and charging coordination	Reliable operation demonstrated in integrated pilots
Network-scale performance	Response time, service-delivery accuracy, delivered power, service duration, and grid-code compliance	Frequency regulation: response time and delivery accuracy; peak shaving: delivered power, service duration, and feeder-load reduction	Sustained grid support without unacceptable network or mobility impacts
System readiness	Certification, market access, settlement procedures, warranties, and distribution of costs and benefits	Commercial frequency regulation and peak shaving: market eligibility and owner compensation	Technical, commercial, regulatory, and social viability demonstrated at the intended deployment scale

Note: Criteria and performance thresholds are service- and context-dependent. The examples are illustrative and should not be interpreted as universal requirements. Advancement between deployment gates should be based on criteria defined before testing. These criteria should specify the intended grid service, operating conditions, measurement procedures, acceptable performance limits, and consequences of non-compliance. The proposed gates provide an assessment structure rather than universally validated numerical thresholds. V2G = vehicle-to-grid.

The service-specific examples in Table 5 illustrate how deployment criteria can be translated into practical verification requirements. For frequency regulation, communication latency, communication availability, response time, and delivered capacity are important indicators, whereas peak shaving requires assessment of delivered power, service duration, and feeder-load reduction. Renewable-energy integration requires evidence of coordinated charging and vehicle availability. These criteria are context-dependent rather than universal, and progression between deployment stages should occur only when the relevant indicators meet predefined requirements for the intended service, network, market, and participant setting.

5.6 Limitations of the Review

This review has several limitations. First, as a critical integrative review, it prioritizes analytical breadth and cross-layer synthesis rather than exhaustive systematic coverage; therefore, the evidence base should not be interpreted as a complete inventory of the V2G literature. Second, substantial heterogeneity in system boundaries, baselines, network models, EV penetration levels, mobility assumptions, battery-aging formulations, performance metrics, and validation environments prevents statistical pooling and limits direct numerical comparisons. Third, a substantial proportion of the available evidence is derived from simulations, laboratory testing, or context-specific demonstrations, limiting the generalizability of findings to heterogeneous fleets, multi-vendor systems, diverse markets, and sustained real-world operation. Fourth, the proposed capability levels and evidence gates are conceptual, service-dependent constructs rather than universally validated thresholds. To avoid overinterpretation, this review treats quantitative findings as study-specific and assesses them in relation to their underlying assumptions, system boundaries, and validation maturity. Future research should operationalize and calibrate the proposed framework using evidence from multi-site field studies across different grid services, fleet compositions, market arrangements, and regulatory settings.

6 Conclusions

This review finds that smart charging and V2G technology can provide valuable flexibility to power systems, but their contribution depends on dependable service capability rather than installed battery capacity or isolated controller performance. This review's central contribution is an integrated, evidence-gated framework that links

operational qualification of installed, connected, controllable, and contractually dependable fleet capacity to the maturity of supporting evidence.

The reviewed literature demonstrates substantial technical potential, particularly in bidirectional power conversion, coordinated charging, peak-demand reduction, renewable-energy utilization, and operating-cost optimization. However, the available evidence remains strongest at the component and simulation levels. Validation involving heterogeneous vehicle fleets, multi-vendor interoperability, cyber-resilient operation, battery-aware compensation, transparent market settlement, and sustained field deployment remains comparatively limited. Consequently, technical feasibility should not be interpreted as evidence of commercial viability or system-level readiness.

The proposed framework offers a structured way to assess whether V2G systems are ready to provide reliable services under actual operating conditions. For sustainable energy systems, this assessment must account for renewable-energy integration, distribution-network constraints, battery degradation, and the costs borne by participating vehicle owners. Further work should test the framework through multi-site pilots and long-term field studies involving different grid services, vehicle fleets, and electricity markets.

Author Contributions

Conceptualization, S.H. and M.N.H.; methodology, S.H. and M.N.H.; investigation, S.H. and M.N.H.; formal analysis, M.I.M.; writing—original draft preparation, S.H. and M.N.H.; writing—review and editing, S.H., M.N.H., M.I.M., and W.T.K.; supervision, H.M.C. All authors have read and agreed to the published version of the manuscript, Shuvo Hossain and Md. Naem Hossain contributed equally to this work and share the first authorship.

Data Availability

Not applicable.

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Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this work, the authors used ChatGPT (OpenAI) to improve the language and readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Nomenclature

AC	Alternating current
AI	Artificial intelligence
BMS	Battery management system
DC	Direct current
EV	Electric vehicle
EVSE	Electric vehicle supply equipment
G2V	Grid-to-vehicle
HIL	Hardware-in-the-loop
IEC	International Electrotechnical Commission
IoV	Internet of Vehicles
ISO	International Organization for Standardization
PV	Photovoltaic

RL	Reinforcement learning
SEI	Solid electrolyte interphase
SoC	State of charge
THD	Total harmonic distortion
TOU	Time-of-use
V1G	Managed unidirectional charging
V2G	Vehicle-to-grid
VPP	Virtual power plant